

Manuely Feliz Portes

Ithaca, NY 14853 | (787) 409-1131 | [LinkedIn](#)
mf833@cornell.edu

EDUCATION

Cornell University, College of Engineering, Ithaca, NY
Bachelor of Science, Mechanical Engineer

Expected May 2027

Hispanic Scholar(HSF), National Women in Computing Award

Relevant Courses: Statics, Introduction to Mechanical Design, Linear Algebra, Introduction to Python, Fluids Mechanics*, System Dynamics*, Mechanics of Materials*, Mechatronics* (*Expected Spring 2025)

PROFESSIONAL EXPERIENCE

Combat Robotics @ Cornell – Member | Ithaca, NY

Aug 2023 – Present

- Collaborated with a team of 10 engineers to design, manufacture, assemble, and test 12-pound combat robots (low- and high-kinetic energy) for the National Havoc Robot League.
- Designed and tested components using Fusion 360 and Finite Element Analysis (FEA) to identify failure points.
- Performed 5+ torque, drag, and gyroscopic calculations to optimize weapon speed up to 180 mph.
- Manufactured 3+ custom parts using lathe and milling machines.

Purdue University – Plant Sciences Research Assistant | West Lafayette, IN

May 2025 – Aug 2025

- Developed an algorithm with 90%+ accuracy to calculate Leaf Area Index (LAI) from UAV LiDAR datasets using gap fraction analysis and optimized filtering.
- Processed and analyzed multisensor UAV data (LiDAR and RGB) against real-world cornfield measurements.
- Compiled technical reports and presented results to senior management.

Testbox Project, Combat Robotics @ Cornell – Member

Jan 2025 – May 2025

- Welded a 6-piece portable steel frame to test combat robots safely.
- Cut and assembled wood panels into a reinforced 3-piece floor capable of withstanding high-energy collisions.
- Integrated modular features, including wheels, to enhance portability and ease of maintenance.

Google Code Next – Engineer | Remote

Aug 2022 – Jul 2023

- Developed 6 programming projects in HTML, Java, and Python.
- Presented projects to 3+ mentors and management using Microsoft PowerPoint.

LEADERSHIP EXPERIENCE

Combat Robotics @ Cornell – Subteam Lead | Ithaca, NY

Aug 2024 – Aug 2025

- Directed a subteam of 9 engineers to design, manufacture, and assemble a 12-pound lifter weapon robot.
- Managed a 16-week agenda timeline, ensuring schedules were met and the budget was maintained under \$6,000.

Society of Hispanic Engineers – Academic Chair | Cornell University

Aug 2024 – May 2025

- Coordinated 3 professional development events, including resume workshops and cultural awareness sessions.
- Built a supportive community of 30+ engineers, fostering inclusion and representation of Hispanic students in STEM.

PROJECT EXPERIENCE

Product Development – Cornell University

Jan 2025 – May 2025

- Designed and tested a portable attachable desk within a \$500 budget for classroom use.
- Iterated design improvements through 3 testing cycles and presented the final product to 3 judges.

SPECIALIZED SKILLS

Programs: AutoCAD, MATLAB, Java, C++, Python, HTML, Excel, PowerPoint, Word

Languages: Spanish (fluent)

Manufacturing: Mill, Lathe, 3D Printing(Bambu), Welding, PowerTools, Band and Chop Saws, Soldering